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CSCI E-102 Econometrics and Causal Inference with R
Part I of Assignment 2

We want to design a survey that uses a sample of independent identically distributed (i.i.d.) observations to estimate the proportion p of families of a large population who are living below the poverty level. Suppose we need the standard error, $SE(\hat{p})$, of the estimate to be no more than 0.002 (0.2%). In each case below, find the necessary sample size.

(a) It is known that this proportion p is roughly 0.11 (11%).

(b) We want to make no prior assumptions about how large this proportion is, i.e., the standard error should be no more than 0.002, regardless of the true proportion value.

SOLUTION:

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$$SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

$p = 0.11$ $\quad \hat{p} = 0.002$ $\quad n = \text{unknown}$

$$0.002 = \sqrt{\frac{0.11(1-0.11)}{n}}$$

$$0.002 = \sqrt{\frac{0.11(0.89)}{n}}$$

$$0.002 = \sqrt{\frac{0.0979}{n}}$$

$$0.002^2 = \frac{0.0979}{n}$$

$$0.000004 = \frac{0.0979}{n}$$

$$\frac{0.000004(n)}{0.000004} = \frac{0.0979}{0.000004}$$

$$n = 24{,}475$$